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# Earth's Sky: The Brightest Stars

The 20 Luminaries of the Night Sky

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## Introduction

On any clear dark night, far from the glow of cities, a person with ordinary eyesight can see about four and a half thousand stars. Almost all of them are faint, and a few dozen stand out: a band of brilliant beacons, white and gold and orange and blue, scattered around the whole vault of the sky. Those few dozen are the subject of this book. They are the stars that every civilization has named first, the stars that steered ships and timed plantings, the stars that became the fixed points of myth and the calibration points of science. They are, in the plainest sense, the most important stars there are, because they are the ones humanity actually watched.

It is worth saying at the start what "brightest" means, because the word hides a subtlety that runs through every chapter that follows. Brightness in the sky, apparent brightness, is a mixture of two very different things: how luminous a star truly is, and how far away it happens to be. Sirius outshines every other star in the night sky, yet it is a fairly ordinary main-sequence star, twenty-five times as luminous as the Sun, distinguished mainly by sitting a mere eight and a half light-years away, the fifth nearest star system to the Sun. Deneb, by contrast, is barely a first-magnitude star, well down the brightness parade, yet it pours out something like a hundred thousand times the Sun's light and lies more than a thousand light-years distant, farther than every other star visible to the naked eye. The brightest stars in our sky are, mostly, our neighbors: a few genuinely mighty beacons, and a larger number of modest stars that simply live next door.

The story of the bright stars is therefore three stories braided together. It is a story of the stars themselves: what they are, how they work, why they shine in the colors they do, and how the brightest among them divide into dwarfs and giants, singles and multiples, steady lights and variables. It is a story of measurement: the magnitude scale, inherited from the ancient astronomers and refined into a precise logarithmic instrument, the photometers and space telescopes that turned the eye's impression into numbers, and the distances, parallax first and cleverness afterward, that let us convert brightness into true luminosity. And it is a story of culture: the Sothic calendar of Egypt built on Sirius, the dog days of the Mediterranean summer, Arcturus switching on a world's fair, Canopus steering spacecraft across the solar system, and Betelgeuse keeping half the world waiting for a supernova.

This book follows the braided story in order. It opens with the discovery of the bright stars, in the sense in which every discovery of the sky begins: the naming and ranking of the visible lights by the earliest astronomers. It explains the magnitude scale and the physics behind stellar brightness, the observation and measurement of the bright stars from the naked eye to the Gaia satellite, and the individual characters of the sky's leading lights, star by star, from Sirius down the list. It sets the bright stars in their cultural context, faces the open questions, the distance of Deneb, the fate of Betelgeuse, the mysteries still attached to the very brightest objects in the sky, and closes with the place of these stars in the long history of astronomy and in the sky of the future.

A note on how to use the numbers in this book, because stellar astronomy is a subject of refinements and the reader will meet slightly different figures elsewhere. Distances, luminosities, and temperatures for the bright stars are known with high but not perfect precision: parallax missions keep improving the distances, the luminosities of the farthest supergiants move with their distances, and the temperatures shift by small recalibrations of the color laws. The figures quoted here are the standard modern values, rounded honestly, and the small print of the trade, the error bars, is itself part of the story, told in the chapters on measurement and on the open questions. What does not change is the structure of the subject: the scale, the inverse-square law, the two populations of the bright-star list, and the individual characters of the stars themselves, and the reader who holds the structure can update any number without losing the book.

A reader's map of what follows may help, since the braided story crosses disciplines that usually live in different books. The physical chapters, two through six, explain the magnitude scale, the physics, the measurement, and the individual stars, and can be read in order or used as reference; the cultural chapters, one and seven, carry the naming and the human contexts; the frontier chapters, eight and nine, carry the open questions and the lighter lore, and the tenth chapter and the conclusion gather the meaning. Nothing in the book assumes prior astronomy beyond a clear night's acquaintance, and the technical vocabulary, magnitude, luminosity, parallax, spectral class, is defined at first use and reused honestly; the reader who wants the numbers carried further will find them in the standard catalogs, and the reader who wants only the view will find the view still works, five thousand years after the first list of bright stars was pressed into clay.

A last preliminary: the Sun is not in this book. It is, of course, the brightest star in the Earth's sky by a factor of

billions, but it belongs to the literature of the Sun and of the solar system, and the stars this book follows are the ones that share the night with it, the fixed beacons of the old celestial sphere. The Sun's place in the story is only as the standard candle of the trade: every luminosity in these pages is given against it, and every neighbor in the brightness parade is, in truth, another Sun, at another distance, in another stage of its life.

A word on the sky of the whole book: the bright-star list is drawn from the whole celestial sphere, and no single place on Earth sees all of it. The northern observer gets the winter triangle and the summer triangle and the pole's attendants, the southern observer gets Canopus and the Cross and the Pointers, and the tropical latitudes, in the best of the tradition's ironies, get nearly everything. The book's star-by-star walk therefore names stars that some readers have never seen, and that is part of the subject too: the bright-star list is one of the last natural inheritances that still varies with latitude, a sky that changes under the traveler's feet, and the reader who moves north or south by a few thousand kilometers will find the parade rearranged, bright members lost and new ones risen, as the oldest geography lesson there is.

## Chapter 1: The Discovery of The Brightest Stars

No one ever discovered these particular lights; they were given names. The distinction matters, because it marks the bright stars off from almost every other object in astronomy. Galaxies, planets beyond Jupiter, nebulae, and most variable stars had to be found, often by instrument and often late; the brightest stars were the first objects any human ever studied, present in every sky our species has ever looked at, and the record of their naming is among the oldest records we have. The name Sirius itself, and the names of Arcturus, Capella, and a few others, are generally traced to roots older than the earliest written languages, which means that when we speak the names of the brightest stars we are using some of the oldest continuous words in existence.

The oldest organized evidence comes from the Mesopotamian and Egyptian traditions. Babylonian star lists, and the omen series built on them, name the principal stars and track their risings with the calendar; Egypt went further and built its civil life on one of them. The hieroglyph of the star Spdt, Sothis, the fairest of all star names' subjects, denotes Sirius, and the heliacal rising of Sirius, its first dawn appearance after the weeks of invisibility, coincided in the third millennium with the annual flood of the Nile, and served as the astronomical alarm of the Egyptian year. The coincidence was so useful that Egypt tracked the star's cycle, and the slow drift of the calendar against it, for over a millennium: a national calendar anchored to the brightest of the fixed stars.

The Greek astronomers organized what they inherited. The constellational star maps of the Hellenistic world, culminating in Ptolemy's great catalog, assigned each bright star a place, a description, and a brightness class, and it is in this catalog that the concept which still governs stellar astronomy, the magnitude scale, makes its formal entrance. Ptolemy, following Hipparchus, ranked the stars into six classes: first magnitude for the brightest, sixth for the faintest visible, and the classes were defined by the ancient word for the greatest, mezon, for the brightest stars. It is a peculiarity of the science's history that the brightest stars carry the smallest numbers, and the reason is simply that the modern scale inherited the ancient one: a Sirius at minus one and a half is brighter than first magnitude by half a class, and the scale had to grow downward when the astronomers of the telescope age measured it.

The Arabic tradition carried the names forward and enriched them. The Islamic astronomers of the medieval period translated and extended the Greek catalogs, and the star names that survived into modern usage, Aldebaran, Altair, Deneb, Fomalhaut, Rigel, Betelgeuse, Vega, and many more, are largely Arabic, descriptive words and phrases that the European star charts of the Renaissance adopted through the Latin translations. The bright stars therefore carry a triple ancestry: prehistoric sounds, Greek geometry, and Arabic description, and a modern observer who says that Aldebaran follows the Pleiades across the sky is speaking three dead languages in one sentence.

The age of the telescope changed the bright stars' meaning more than their roster. Until the seventeenth century, the bright stars were the sky's principal contents, and the faint stars were decoration; after Galileo, the roles reversed, and every addition of optical power multiplied the faint multitude while the bright list stayed fixed at a few dozen. The discoveries at the bright stars came to be of a new kind: not the stars themselves, but their companions and their facts. In 1718 Edmond Halley showed that the bright stars move, detecting the proper motion of Arcturus and Sirius against the catalog positions of Ptolemy; in 1838 Friedrich Bessel measured the parallax of the double bright star 61 Cygni and a few months later of Sirius's neighbor, the first distances of stars; in 1844 Bessel read the wobble of Sirius and Procyon and inferred invisible companions, and in 1862 the telescope-maker Alvan Clark, testing a new lens, actually found Sirius B, the first known white dwarf. The brightest

stars, because they were the nearest and the easiest to measure, led the way into stellar astronomy at every step, and their story through the nineteenth century is the story of the birth of the science of stars itself.

A note belongs here on the man who did for the bright stars' positions what the ancients did for their names: the second-century astronomer Claudius Ptolemy, whose star catalog, published in the work the Arabs called the Almagest, is the foundation document of the modern brightness list. Ptolemy's catalog gives over a thousand stars, each with a position, a constellation, and a magnitude class, and its first-magnitude entries, Sirius, Canopus, Arcturus, Vega, Capella, and their peers, are the same stars that head the modern list; the catalog was copied, translated, and used for fourteen centuries, and the magnitude classes it assigned are the direct ancestors of the numbers the satellites now measure. Historians have argued for generations over how much of the catalog is Hipparchus, working three centuries earlier, and how much is Ptolemy's own observation, and the argument is itself part of the bright stars' discovery story: the oldest brightness ranking in existence is also one of the most studied texts in the history of science, and every modern magnitude carries its two-thousand-year pedigree back to those six classes.

One more discovery of the bright stars deserves its place, because it closed the ancient ranking and opened the modern one: the measurement of the scale's steps. The ancient classes, first through sixth magnitude, were equal steps of appearance; the question of the nineteenth century was whether the steps were equal in fact, and the photometers answered that they were, to the approximation that made Pogson's ratio the law of the scale. The measurement, made on the brightest stars because they were the standard case, converted a ranking into a logarithmic quantity and a tradition into an instrument, and the instrument then ran past its founders: the scale extended upward to negative magnitudes for the sky's brightest and downward through the telescope's fainter and fainter fields, until the same six steps of the ancient eye covered forty magnitudes of the modern census, a brightness range of a hundred billion to one, all on the scale the Dog Star heads. The discovery of the bright stars, told as measurement rather than naming, ends with that arithmetic: the sky ranked by eye, two thousand years, then ranked by ratio, exactly, in a single century.

A final discovery of the bright-star story belongs to the space age and completes the arc of this chapter: the measurement of the whole list from orbit. The Hipparcos satellite, launched in 1989, measured the positions, parallaxes, proper motions, and brightnesses of a hundred thousand stars and more, including every star the eye can see, to a precision the ground had never approached; the Tycho catalogs extended the coverage to more than a million; and the Gaia mission, a generation later, multiplied the census past a billion and fixed the bright stars' distances to a fraction of a percent. The brightest stars, the first objects of the ancient catalogs, are therefore also the best-measured objects of the modern ones, and the discovery of the bright stars ends where it began, with a complete and ordered list of the sky's brightest lights: pressed into clay in the Bronze Age and delivered as a table of milliarcsecond astrometry in the twenty-first, the same sky, the same names, and the same short parade of brilliant stars, measured now to the limits of the arithmetic.

The twentieth century completed the roster by fixing the scale that ranks them. The visual magnitude scale was calibrated by the astronomers of the photometric era, anchored to the measured brightnesses of the standard stars, and extended in both directions, to negative magnitudes above Sirius and to faintness far below the naked eye, and the bright-star list settled into its modern order: Sirius first, Canopus second, the system of Alpha Centauri third, Arcturus fourth, Vega fifth, Capella sixth, Rigel seventh, Procyon eighth, Achernar ninth, Hadar tenth, Altair eleventh, Acrux twelfth. The order is now known to thousandths of a magnitude, and the list, checked by instruments in space that see the whole sky without an atmosphere, is closed at the top. The discovery of the bright stars, which began with the first words ever spoken about the sky, ended, in effect, with the Hipparcos catalog of the 1990s, and what remains is the deeper work this book is mostly about: understanding what the bright stars are.